Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A siren emits 500 Hz steadily while approaching a stationary listener. Compare observed and emitted frequency and explain.

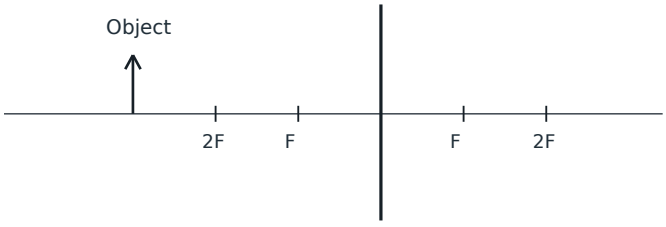
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $1/f = 1/d_o + 1/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is approximately undeviated in the thin-lens model.

Your turn: Draw these two rays and mark their intersection for an object beyond 2 f.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Complete ray sketches for parallel rays through a converging lens and a diverging lens. Label the optical axis 0.

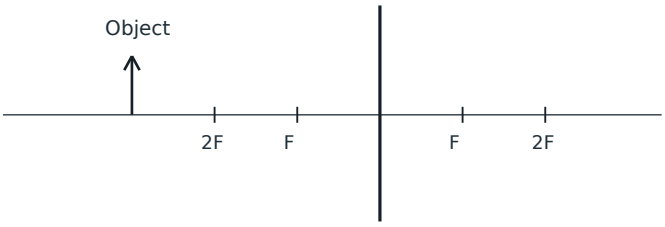


Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

04 Apply the physics

A small lamp sits 3f from a converging lens, as marked. Draw a ray parallel to the axis and a ray through the lens center. Use their intersection to classify the image as upright/inverted and real/virtual.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A magnifying glass produces an upright virtual image when object is inside f. Explain why “converging lens always makes a real image” is wrong.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave has period 4 s. Find the cycles completed in 24 s and its frequency.
Label a wave sketch to support the calculation or comparison.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

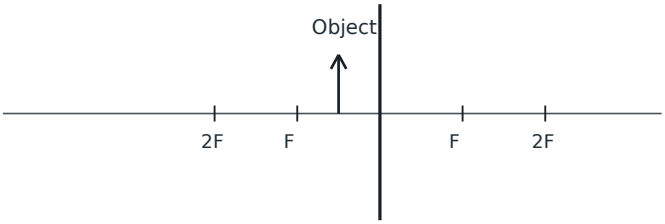
Which lens spreads parallel rays: converging or diverging?



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Move the object between the converging lens and its focal point. Predict whether outgoing rays meet physically beyond the lens and describe the image qualitatively.



Converging thin lens; add principal rays.

Marzano check - circle: 1 2 3